STEM Fellowship 2025 Inter-University Health Data and AI Inquiry Program

Predicting Chronic Illness in Aging Populations Using Physiological Markers

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Abstract

As populations age, early detection of chronic illness is increasingly vital to prevent declines in independence and quality of life. This study investigates whether wearable-derived physiological signals and clinical health records can be used to predict chronic illness in older adults using machine learning models.

We used data from the All of Us (AOU) Research Program, including Fitbit wearable summaries and EHR-based clinical measures, to predict three high-burden chronic conditions: osteoporosis, dementia, and heart failure. After filtering for data completeness, we created participant-level summaries using 21 features spanning physical activity, sleep behavior, cardiovascular metrics, and biometric indicators. A final dataset of approximately 2,000 participants aged 60+ was used for modeling.

Logistic regression, random forest, and XGBoost models were trained and evaluated using 5-fold cross-validation with stratified train/test splits. We addressed class imbalance with upsampling during training. Performance was measured using accuracy, F1-score, and area under the ROC curve (AUC).

Osteoporosis prediction achieved the highest performance across models (XGBoost AUC = 0.802, F1 = 0.742), supported by features such as basal metabolic rate (BMR) and daily energy expenditure. Dementia and heart failure results were limited by severe class imbalance, with undefined F1-scores and reduced AUCs.

These results demonstrate the feasibility of predicting musculoskeletal chronic conditions using wearable and clinical data, and highlight the challenges of modeling rare disease outcomes in imbalanced datasets. Our findings suggest a scalable framework for early risk stratification using passive health monitoring in aging populations.

Keywords
chronic disease, machine learning, wearable data, aging, health prediction, All of Us

1 Introduction

Chronic illness is a leading cause of disability and healthcare utilization in older adults. In the United States, over 92% of individuals aged 60+ live with at least one chronic condition, and nearly 80% manage two or more [1]. Early detection is critical, as chronic disease often marks the transition from functional independence to long-term care.

Machine learning (ML) offers a promising path for identifying subtle physiological changes that precede clinical diagnosis. Prior studies using electronic health records (EHRs) have shown ML models can achieve predictive accuracies of over 80% within one-month diagnosis windows [2]. In parallel, wearable devices like Fitbit trackers provide a continuous, non-invasive stream of behavioral and physiological data that can enhance predictive modeling [3].

In this study, we use data from the All of Us (AOU) Research Program, combining wearable summaries and EHR-based clinical measurements to train ML models that predict three chronic conditions in older adults: **osteoporosis**, **dementia**, and **heart failure**. These conditions span musculoskeletal, neurocognitive, and cardiovascular systems and are prevalent, progressive, and impactful in geriatric care.

We focus on modifiable, real-world health signals—such as physical activity, sleep quality, and heart rate variability—to explore whether passive monitoring paired with ML can enable earlier risk detection and inform scalable prevention strategies in aging populations.

2 Materials & Methods

We used data from the *All of Us* (AOU) Research Program, a nationwide NIH initiative that provides structured access to electronic health records (EHRs), clinical measurements, and wearable device data, including Fitbit summaries. Our analysis focused on participants aged 65 and older who had complete Fitbit and clinical data available. The demographic makeup of AOU supports this focus: 44.4% of participants are over age 60, and 42.4% identify as non-white.

Chronic illness status was labeled using OMOP concept IDs from the AOU `condition_occurrence` table. We modeled three prevalent, age-associated conditions: osteoporosis (80502), dementia (4182210), and heart failure (316139). Binary flags for each diagnosis were assigned per participant.

To construct predictors, we extracted 21 physiological features from three Fitbit domains and AOU clinical tables:

- **Physical activity and energy expenditure:** daily step count, calories burned, basal metabolic rate (BMR), floors climbed, sedentary and active minutes, elevation gain
- **Cardiovascular function:** minimum/maximum heart rate, calories from heart rate zones, systolic/diastolic blood pressure, total minutes in heart rate zones
- **Sleep behavior:** total sleep, REM, light, deep, and wake durations
- **Clinical metrics:** BMI, body weight

We merged Fitbit and clinical tables using the shared `person_id` key. To create a consistent, participant-level dataset, repeated measurements (e.g., daily steps or nightly sleep) were averaged per individual. Clinical variables were reshaped and merged with wearable summaries. Participants with missing values in any predictor column were excluded, yielding a final cohort of approximately 2,000 older adults.

For modeling, we trained two sophisticated machine learning models from built-in R packages (caret, pROC, xgboost, randomForest).

- An ensemble learning technique called **Random Forest** creates a large number of decision trees and combines their results to get a forecast that is more reliable. It resists overfitting and performs well with nonlinear linkages and interactions.
- **Extreme Gradient Boosting**, or XGBoost, is a potent tree-based method that excels in processing structured (tabular) data quickly and accurately. By training progressively and minimizing error at each stage, XGBoost outperforms conventional decision trees and is hence well-suited for high-dimensional, mixed-type data.

We divided the dataset into 75% training and 25% testing subsets in order to assess generalization, or how well the model works on fresh, untested data. To avoid overfitting and maximize hyperparameters, we used 5-fold cross-validation when training models on the training set. To guarantee consistency, cross-validation splits the training data into five sections, trains the model on four of them, and then validates it on the fifth.

Three metrics were used to evaluate the model's performance:

- **Accuracy:** the proportion of correct predictions made by the model. Despite being straightforward, it may be deceptive in datasets that are unbalanced (for example, when one class is far more prevalent than the other).
- The **F1-score** is a balanced indicator of a model's capacity to accurately identify positive cases without over-predicting them. It is calculated as the harmonic mean of precision (% of true positives)and recall (% of correctly identified positives). When there is an imbalance in the classrooms, this is extremely helpful.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Regardless of the decision threshold, the model's ability to differentiate between classes is gauged by its **AUC (Area Under the Receiver Operating Characteristic Curve)**. Perfect discrimination is indicated by an AUC of 1.0; performance that is no better than chance is shown by an AUC of 0.5.

3 Results

For each of the three chronic conditions, osteoporosis, dementia, and heart failure, we assessed the predictive capabilities of three classification algorithms: logistic regression (LM), random forest, and XGBoost. Accuracy, F1-score, and AUC (area under the ROC curve) were used to evaluate the model's performance. We also used the internal feature importance metrics of each model to analyze the top-ranked predictors.

Osteoporosis	LM	Rand. Forest	XGBoost
Accuracy	0.723	0.725	0.745
F1-score	0.704	0.712	0.742
AUC	0.807	0.791	0.802

Table 1: Predictive Performance of Classification Models for Osteoporosis

Dementia	LM	Rand. Forest	XGBoost
Accuracy	0.980	0.989	0.986
F1-Score	Undefined	Undefined	Undefined
AUC	0.686	0.629	0.579

Table 2: Predictive Performance of Classification Models for Dementia

Heart Failure	LM	Rand. Forest	XGBoost
Accuracy	0.902	0.902	0.897
F1-Score	Undefined	Undefined	Undefined
AUC	0.595	0.661	0.678

Table 3: Predictive Performance of Classification Models for Heart Failure

3.1 Osteoporosis

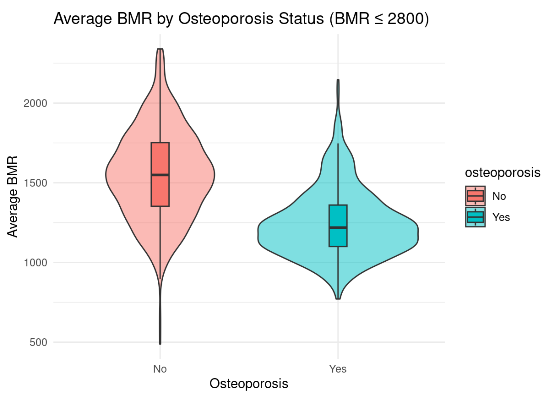

Figure 1: Predictive Performance of Classification Models for Heart Failure

Out of all the models, osteoporosis produced the most reliable and consistent findings. Random forest and logistic regression came in second and third, respectively, to XGBoost, which performed the best with an accuracy of 0.745, F1-score of 0.742, and AUC of 0.802. This implies that the models successfully learned discriminative patterns from clinical and wearable features. Basal metabolic rate (BMR), the average calories burned and the calories obtained from the heart rate data were found to be among the most significant variables by feature importance analysis (Figure 5, top panel). BMR was shown to have a negative association with osteoporosis. These results are consistent with known physiological relationships among musculoskeletal health, energy metabolism, and bone density in aging populations.

3.2 Dementia

Class imbalance made the classification of dementia more difficult. Despite the logistic model's AUC of 0.686 and XGBoost's excellent accuracy of 0.986, all models' F1-scores were undefinable, suggesting that no positive dementia cases in the test set could be recalled. This implies that even with stratified sampling and upsampling during training, the models had trouble generalizing to minority-class situations. However, sleep-related characteristics such as REM sleep, light sleep, and total time in bed were repeatedly found to be important predictors of dementia status by the models (Figure 5, middle panel). These factors show physiological alterations in rest-activity cycles and circadian rhythm, which are known to worsen in the early phases of cognitive decline.

3.3 Heart Failure

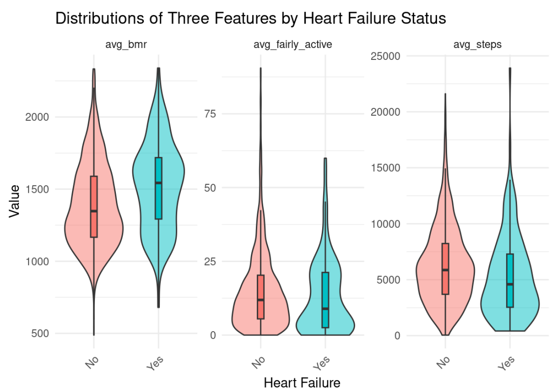

Figure 2: BMR, Activity in minutes and Steps by Heart Failure

The most severe class imbalance affected the prediction of heart failure. AUCs stayed modest, peaking at 0.678 for XGBoost, while F1-scores were not established for any of the models. The infrequent occurrence of heart failure patients in the training set hindered efficient model learning, even with the implementation of stratified test partitioning and upsampling. Activity-related variables, such as fairly active minutes, REM sleep, and heart rate zone lengths, were shown to be the most helpful in differentiating heart failure cases from controls, according to feature importance rankings (Figure 5, bottom panel). Despite the physiological plausibility of these indicators, model performance was unpredictable for this scenario due to a paucity of good training data.

4 Discussion

4.1 Interpretation

All three ML models produced the most reliable results when modeling for osteoporosis diagnosis. Achieving AUCs of around 0.79-0.91 and F1 scores of 0.71-0.74, these models showed strong predictive performance when distinguishing individuals with undiagnosed osteoporosis. The most significant feature across both RF and XGB was the average basal metabolic rate (BMR), followed by BMI and daily calories burned.

These results reflect a UK Biobank analysis of $\tilde{8}00,000$ participants, which demonstrated a correlation between people with higher basal metabolic rates and a lower risk of developing osteoporosis [4]. A similar study of 289 Taiwanese women over the age of 50, using multiple linear regression, found that basal metabolic rate significantly predicted osteoporosis ($p < 0.05$, statistical significance) [5]. In addition, a Chinese cohort study suggests that elevated BMR reflects improved musculoskeletal health and may serve as a modifiable factor to reduce the risk of osteoporosis [6]. Together, these findings validate our ML models' prioritization of average BMR as a feasible indicator of osteoporosis incidence in aging populations.

When modeling for heart failure and dementia, both Random Forest and XGBoost models returned undefined F1 scores. In practice, this results in unreliable classification for identifying positive cases.

4.2 Technical Challenges

The significant class imbalance in our dataset was one of the main obstacles we faced when developing the model. Only a small percentage of the over 60,000 patients in our overall cohort had both a diagnosis of the target chronic conditions and complete wearable data. The final modeling dataset was whittled down to about 2,000 people after being filtered for completeness and combined from various sources; many of these people did not have the target conditions, particularly for heart failure and dementia.

In order to rectify this imbalance, we put a number of mitigating techniques into practice:

- Using the caret framework to upsample the minority class in the training set
- Using stratified train-test splitting to guarantee that both sets contain positive cases
- All predictor variables' median imputation for missing values

These actions only slightly increased model stability under some circumstances, but they were not enough to develop strong decision boundaries in scenarios with significant imbalance. For instance, the positive class in heart failure was so uncommon that there were no "Yes" cases in the training set for a number of runs. Because of this, models were unable to identify any patterns linked to the condition, leading to undefined F1-scores and AUC values below 0.69, which suggest little to no discriminative potential.

A closer look showed that many positive dementia cases slipped disproportionately into the test set, resulting in poor memory and uneven F1 performance across algorithms, even though the model initially produced an apparently excellent AUC of 0.807. This implies that test set imbalance, rather than actual generalization, exaggerated the model's seeming success. Regardless of algorithm complexity, these imbalances restricted our models' capacity to discriminate between participants with and without certain conditions. The underrepresentation of important clinical classes in the training data ultimately limited model performance, even when high-capacity models like XGBoost were used.

When applied to targets with appropriate power, osteoporosis, the only condition with fairly balanced representation, achieved robust and consistent results across all parameters, confirming the validity of our whole pipeline. Reflecting established physiological links with aging and chronic illness, the most significant predictors across models were fairly active minutes, REM/light sleep duration, and basal metabolic rate (BMR).

4.3 Implications

4.3.1 Fitbit

Consumer wearables such as Fitbit have reached mass adoption, with one in five American adults regularly wearing a fitness tracker or smartwatch, according to a 2019 Pew Research Center Survey [7]. The high-volume data generated through wearables offers tremendous potential for training ML models to detect patterns and accurately predict health outcomes, including early disease diagnosis. Particularly, in our osteoporosis models, Fitbit-derived metrics such as average basal metabolic rate (BMR) demonstrated strong predictive performances for ML algorithms.

4.3.2 Importance of Preventing Osteoporosis

Osteoporosis is a degenerative disease which causes loss of bone density, leaving victims more susceptible to fractures. Among older adults, bone fractures can cause decreased mobility, chronic pain, loss of functional independence, and even early deaths. Osteoporosis is common in older adults, with "in 2017–2018, the age-adjusted prevalence of osteoporosis at either the femur neck or lumbar spine or both among adults aged 50 and over was 12.6% and was higher among women (19.6%) compared with men (4.4%)." [8]. Early identification of at-risk individuals allows for timely interventions, which can significantly delay or prevent the onset of severe bone loss.

5 Conclusions

This study demonstrates that integrating wearable-derived metrics with electronic health records introduces the potential for reliable detection of age-related chronic diseases. As demonstrated by our Random Forest and XGBoost ML models, Fitbit-generated features such as average basal metabolic rate (BMR) can achieve robust predictive performance(AUC 0.80 and Accuracy of 74.5%) when predicting for osteoporosis. Although severe class imbalance limited dementia and heart-failure modeling, the strong results for osteoporosis demonstrate that with larger or better-balanced cohorts, comparable predictive accuracy could be achieved for other chronic diseases.

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This study used data from the All of Us Research Program's Controlled Tier Dataset v8, available to authorized users on the Researcher Workbench.

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6 Reproducibility Statement

All code used for data cleaning, modeling, and analysis is available at: https://github.com/cybwifey/Dream-Team.